Experiment Name: 8-bit Multiplication using Assembly Language

1. Objective

The objective of this experiment is to perform 8-bit multiplication using Assembly Language in the 8086 microprocessor. This experiment helps to understand how multiplication operations are performed using registers and instructions

2. Apparatus / Tools

- Personal Computer (PC)
- EMU8086 Emulator Software

3. Working Procedure

Steps:

1. The first number is loaded into the AL register.
2. The second number is loaded into the BL register.
3. The MUL instruction is used to perform multiplication.
4.The result is stored in the AX register.

4. Program:

```
.MODEL SMALL
.STACK 100H
.DATA
.CODE

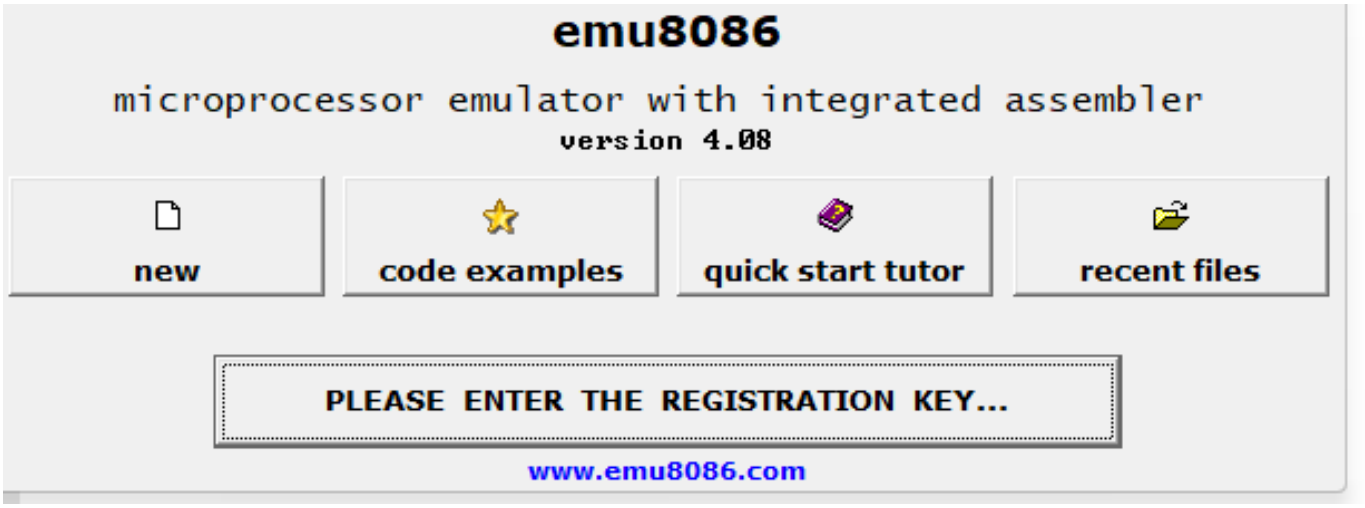
MAIN PROC

  MOV AL,12H
  MOV BL,04H
  MUL BL
  INT 21H

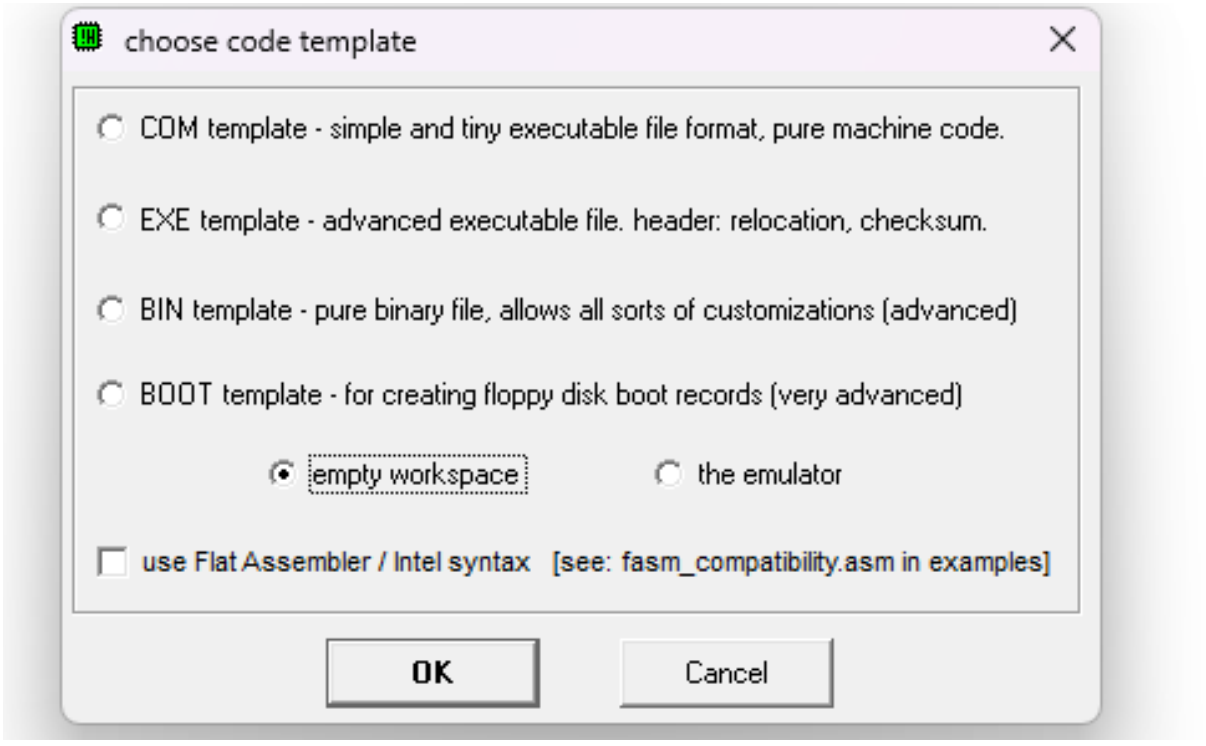
 MAIN ENDP
END MAIN
```

5.Operation Step

Step 1: Open EMU8086 and click on 'New' to start a project.



Step 2: Select “Empty Workspace”
After clicking on “New”, a new window will appear. From there, select “Empty Workspace” and click OK to proceed.



Step 3: Write the Program Code
After opening the workspace, write the assembly language program in the editor window.

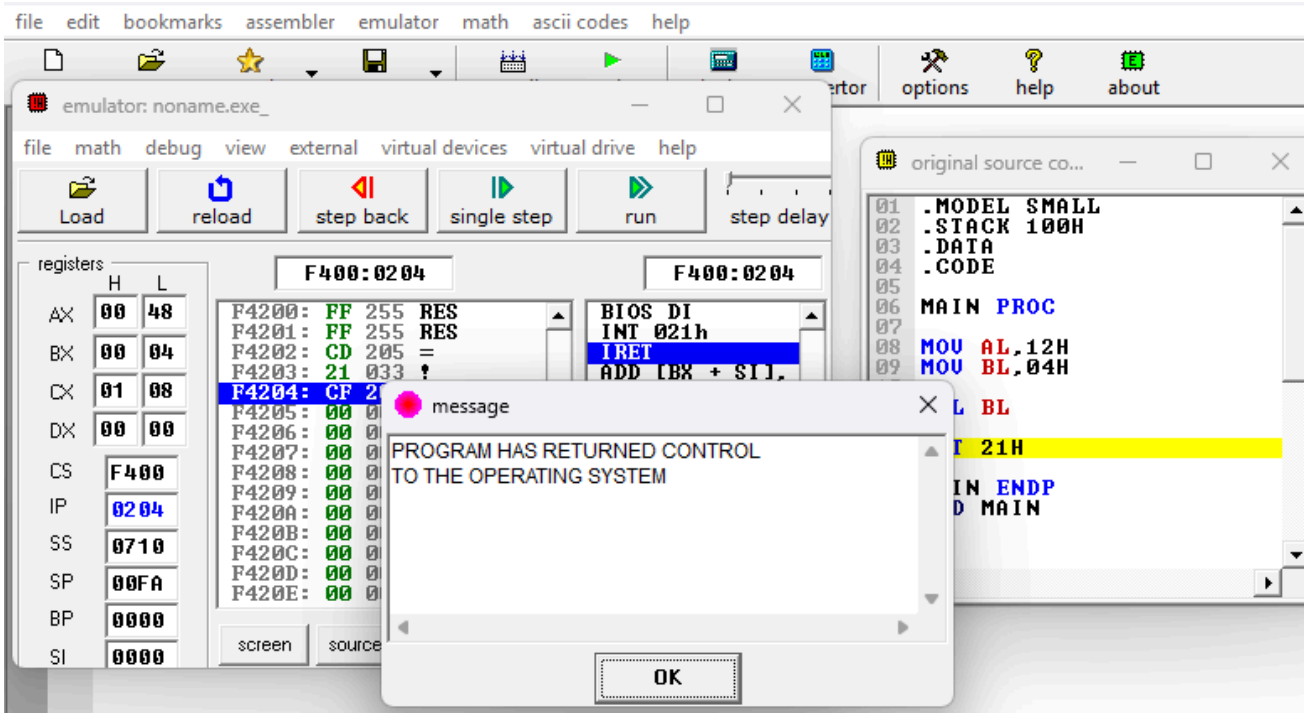
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```
01 .MODEL SMALL
02 .STACK 100H
03 .DATA
04 .CODE
05
06 MAIN PROC
07
08  MOV AL,12H
09  MOV BL,04H
10
11  MUL BL
12
13  INT 21H
14
15 MAIN ENDP
16 END MAIN
```

5.Operation Result

Step 1: Click on the Run/Execute option to run the program. The result of the multiplication will be shown in the AL register.
Result of Multiplication = 48H



6.Comment

The program executed successfully, and the 8-bit multiplication was correctly performed. The result (48H) was stored in the AL register.

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